



Bhabha University, Bhopal (M.P.)

Scheme of Examination

Program – Bachelor Of Technology

Branch – Mechanical Engineering

Session: 2021-2022

Seventh Semester

Sr. No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Marks for End Sem. Examination		Total Marks
			L	T	P		End Sem. Exam.	Tests (Two)	Assign ments /Quiz	End Sem. Practic al /Viva	Practical Record/Assi gnment/Qui z /Presentation	
1	ME-71	Heat and Mass Transfer	3	1	2	5	70	20	10	30	20	150
2	ME-72	Advance Machine Design	2	1	2	4	70	20	10	30	20	150
3	ME-73	Operation Research and Supply Chain	2	1	-	3	70	20	10	-	-	100
4	ME-74	Automobile Engineering	3	-	2	5	70	20	10	30	20	150
5	ME-75	Power Technology	3	-	-	3	70	20	10	-	-	100
6	ME-76	CAD/CAM/CIM	-	-	2	2	-	-	-	30	20	50
7	ME-77	Industrial Training (2 Weeks)	-	-	2	2	-	-	-	30	20	50
		Total	14	3	8	24	350	100	50	150	100	750

L: Lecture T: Tutorial P: Practical

Note: End semester Examination will be of 3 hrs.



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Mechanical Engineering, VII-Semester

ME-71 Heat and Mass Transfer

Course Objectives:

After studying this course, students will be able to

1. Know about the basic concept of heat transfer and its modes.
2. Solve problems based on conduction, convection, and radiation.
3. Differentiate the modes of heat transfer i.e. conduction, convection, and radiation
4. Understand the working principle and types of heat exchangers.
5. Understand the concept of boiling and condensation, mass transfer.

Syllabus:

Unit-1 Basic Concepts: Modes of heat transfer, Fourier's law, Newton's law, Stefan Boltzman law; thermal resistance and conductance, analogy between flow of heat and electricity, combined heat transfer process; Conduction: Fourier heat conduction equation, its form in rectangular, cylindrical and spherical coordinates, thermal diffusivity, linear one dimensional steady state conduction through a slab, tubes, spherical shells and composite structures, electrical analogies, critical insulation- thickness for pipes, effect of variable thermal conductivity.

Unit 2 Extended Surfaces (fins): Heat transfer from a straight and annular fin (plate) for a uniform cross section; error in measurement of temperature in a thermometer well, fin efficiency, fin effectiveness, applications; Unsteady heat conduction: Transient and periodic conduction, heating and cooling of bodies with known temperatures distribution, systems with infinite thermal conductivity, response of thermocouples.

Unit 3 Convection: Introduction, free and forced convection; principle of dimensional analysis, Buckingham 'pie' theorem, application of dimensional analysis of free and forced convection, empirical correlations for laminar and turbulent flow over flat plate and tubular geometry; calculation of convective heat transfer coefficient using data book.

Unit 4 Heat Exchangers: Types- parallel flow, counter flow; evaporator and condensers, overall heat transfers coefficient, fouling factors, log-mean temperature difference (LMTD), method of heat exchanger analysis, effectiveness of heat exchanger, NTU method;

Mass transfer: Fick's law, equi-molar diffusion, diffusion coefficient, analogy with heat transfer, diffusion of vapour in a stationary medium.



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Mechanical Engineering, VII-Semester

ME-71 Heat and Mass Transfer

Unit 5 Thermal Radiation : Nature of radiation, emissive power, absorption, transmission, reflection and emission of radiation, Planck's distribution law, radiation from real surfaces; radiation
heat exchange between black and gray surfaces, shape factor, analogical electrical network, radiation shields.

Boiling and condensation: Film wise and drop wise condensation; Nusselt theory for film wise condensation on a vertical plate and its modification for horizontal tubes; boiling heat transfer phenomenon, regimes of boiling, boiling correlations.

References:

1. Sukhatme SP; Heat and mass transfer; University Press Hyderabad
2. Holman JP; Heat transfer; TMH
3. Nag PK; heat and Mass Transfer; TMH
4. Domkundwar, Heat and Mass Transfer, Dhanpt Rai & Co.
5. Sachdeva R.C., Fundamentals of Engineering Heat and Mass Transfer, New Age Science
6. Dutta BK; Heat Transfer Principles And App; PHI Learning
7. Mills AF and Ganesan V; Heat transfer; Pearson
8. Cengel Yunus A; Heat and Mass transfer; TMH
9. Yadav R; Heat and Mass Transfer; Central India pub-Allahabad
10. Incropera FP and Dewitt DP; Heat and Mass transfer; Wiley



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Mechanical Engineering, VII-Semester

ME-71 Heat and Mass Transfer

List of Experiments :

1. Conduction through a rod to determine thermal conductivity of material
2. Forced and free convection over circular cylinder
3. Free convection from extended surfaces
4. Parallel flow and counter flow heat exchanger effectiveness and heat transfer rate
5. Calibration of thermocouple
- 6 . Experimental determination of Stefan-Boltzman constant



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Mechanical Engineering, VII-Semester

ME-72 Advance Machine Design

Course Objective

Understand the design concepts of belt, rope and chain drives.
Able to design different types of gears.
Able to design I.C. Engine components, different types of couplings and power screw
Inspect miscellaneous components such as flanged coupling, rigid coupling, and pressure vessels.

Course Contents:

Unit- I: Design of Belt, Rope and Chain Drives: Methods of power transmission, design of flat belt drive and V-belt drive ; Design of chain drives, roller chain and its selection; Design of rope drives.

Unit- II: Spur and Helical Gears: Force analysis of gear tooth, AGMA Bending stress equation and AGMA Contact stress equation, modes of failure, beam strength, Lewis equation, form factor, formative gear and virtual number of teeth; Gear materials; Surface strength and wear of teeth; strength against wear; Design of straight tooth spur and Helical Gears.

Unit- III: Bevel Gears: Application of bevel, formative gear and virtual number of teeth; Force analysis; Lewis equation for bevel gears; Strength against wear; Design of bevel gear.

Unit- IV: Design of I.C. Engine Components: General design considerations in I C engines; design of cylinder; design of piston and piston-rings; design of connecting rod; design of crankshaft.

Unit -V: Design of Miscellaneous Components: Design of Knuckle joint, Design of Cotter joint, Design of keys, Design of Flanged coupling; Rigid coupling and Flexible coupling ,Design of Pressure vessels subjected to internal pressure, Design of power screw.



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Mechanical Engineering, VII-Semester

ME-72 Advance Machine Design

References:

1. Shigley J.E.; Machine Design; TMH
2. Bhandari VB; Design of Machine Elements; TMH
3. Abdul Mubeen; Machine Design; Khanna Publishers
4. Sharma & Agrawal; Machine Design; Katson
4. Sharma CS and Purohit K; Design of Machine Elements; PHI Learning.
5. Dwivedi and Pandey; Machine Drawing and Design, Dhanpat Rai & Co.
6. Wentzell TH; Machine Design; Cengage Learning
7. Hall and Somani; Machine Design; Schaum Series; TMH
8. Kulkarni SG; Machine Design; TMH
9. Norton R; Design Of Machinery; TMH

Note: PSG Design data book and/ or Mahadevan and Reddy's Mechanical design data book are to be provided/ permitted in exam hall (duly verified by authority)

List of Experiment (Pl. expand it):

Designing and sketching of components contained in the syllabus



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Mechanical Engineering, VII-Semester

ME-73 Operation Research and Supply Chain

Unit 1 Linear system and distribution models: Mathematical formulation of linear systems by LP, solution of LP for two variables only, special cases of transportation and assignment and its solution, Vogel's forward looking penalty method, cell evaluation degeneracy, use of SW Lindo, Tora, Excell.

Unit 2 Supply chain (SCM): Definition, importance, expenditure and opportunities in SCM; integration of inbound, outbound logistics and manufacturing to SCM, flow of material money and information, difficulties in SCM due to local v/s system wide (global) optimization and uncertainties in demand and transportation; Bull-whip effect; customer value; IT, info-sharing and strategic partnerships; plant and warehouse-network configuration; supply contracts and revenue sharing; outsourcing; transportation, cross docking and distribution, forecasting models in SCM; coordination and leadership issues; change of purchasing role and vendor rating, variability from multiple suppliers.

Unit 3 Inventory models: Necessity of inventory in process and safety stock, problem of excess inventory and cycle time ($=WIP/Throughput$), JIT/ lean mfg; basic EOQ/ EPQ models for constant review Q-system(S,s); periodic review, base stock P-system; service level, lead time variance and safety stock;; ABC, VED and other analysis based on shelf life, movement, size, MRP technique and calculations, lot sizing in MRP, linking MRP with JIT; evolution of MRP to ERP to SCM and e-business.

Unit 4(a) Waiting Line Models Introduction, Input process, service mechanism, Queue discipline, single server (M/M/1) average length and times by Little's formula, optimum service rate; basic multiple server models (M/M/s)

(b) Competitive strategy: concept and terminology, assumptions, pure and mixed strategies, zero sum games, saddle point, dominance, graphical, algebraic and LP methods for solving game theory problems.

Unit 5: (a) Decision analysis: decision under certainty, risk probability and uncertainty; Hurwicz criteria; AHP- assigning weight and consistency test of AHP

(b) Meta-heuristics Definition of heuristic and meta-heuristic algorithms; introduction to Tabu search, Simulated Annealing and Genetic algorithms and solution of traveling salesman and non linear optimization problems.



Bhabha University, Bhopal (M.P.)

Mechanical Engineering, VII-Semester

ME-73 Operation Research and Supply Chain

References:

1. Hillier FS and Liberman GJ; Introduction to Operations Research concept and cases; TMH
2. Simchi-Levi, Keminsky; Designing and managing the supply chain; TMH.
3. Srinivasan G; Quantitative Models In Operations and SCM; PHI Learning
4. Mohanty RP and Deshmukh SG; Supply Chain Management; Wiley India
5. Taha H; Operations research; PHI
6. Sen RP; Operations Research-Algorithms and Applications; PHI Learning
7. Sharma JK; Operations Research; Macmillan
8. Ravindran , Philips and Solberg; Operations research; Wiley India
9. Vollman, Berry et al; Manufacturing planning and control for SCM; TMH.
10. Bowersox DJ, Closs DJ, Cooper MB; Supply Chain Logisti Mgt; TMH
11. Burt DN, Dobler DW, StarlingSL; World Class SCM; TMH



Bhabha University, Bhopal (M.P.)

Mechanical Engineering, VII-Semester

ME-74 Automobile Engineering

Unit-I: Chassis & Body Engg: Types, Technical details of commercial vehicles, types of chassis, lay out, types of frames, testing of frames for bending & torsion on unutilized body frame, vehicle body and their construction, driver's visibility and methods for improvement, safety aspects of vehicles, vehicle aerodynamics, optimization of body shape, driver's cab design, body materials, location of engine, front wheel and rear wheel drive, four wheel drive.

Unit-II: Steering System: front axle beam, stub axle, front wheel assembly, principles of types of wheel alignment, front wheel geometry viz. camber, Kingpin inclination, castor, toe-in and toe out, condition for true rolling motion, centre point steering, directional stability of vehicles, steering gear, power steering, slip angle, cornering power, over steer & under steer, gyroscopic effect on steering gears.

Unit-III: Transmission System: Function and types of clutches, single plate, multi-plate clutch, roller & spring clutch, clutch lining and bonding, double declutching, types of gear Boxes, synchroniser, gear materials, determination of gear ratio for vehicles, gear box performance at different vehicle speed, automatic transmission, torque converters, fluid coupling, principle of hydrostatic drive, propeller shaft, constant velocity universal joints, differential gear box, rear axle construction.

Unit-IV: Suspension system : Basic suspension movements, Independent front & rear suspension, shock absorber, type of springs: leaf spring, coil spring, air spring, torsion bar, location of shackles, power calculations, resistance to vehicle motion during acceleration and braking, power & torque curve, torque & mechanical efficiency at different vehicle speeds, weight transfer, braking systems, disc theory, mechanical, hydraulic & pneumatic power brake systems, performance, self-energisation, airbleeding of hydraulic brakes, types of wheels and tyres, tyre specifications, construction and material properties of tyres & tubes.

Unit-V: Electrical and Control Systems: storage battery, construction and operation of lead acid battery, testing of battery, principle of operation of starting mechanism, different drive systems, starter relay switch, regulator electric fuel gauge, fuel pump, horn, wiper, Lighting system, head light dazzling, signaling devices, battery operated vehicles, choppers. importance of maintenance, scheduled and unscheduled maintenance, wheel alignment, trouble Shooting probable causes & remedies of various systems, microprocessor based control system for automobile, intelligent automobile control systems.



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Mechanical Engineering, VII-Semester

ME-74 Automobile Engineering

References:

1. Crouse , Automotive Mechanics TMH.
2. Srinivasan S; Automotive engines; TMH
3. Gupta HN; Internal Combustion Engines; PHI;
4. Joseph Heitner, Automotive Mechanics, Principles and Practices, CBS Pub.
5. Kripal Singh, Automotive Engineering Khanna Pub.
6. Newton & Steeds , Automotive Engineering
7. Emission standards from BIS and Euro –I and Euro-III



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ME-74 Automobile Engineering

List of experiments:

Study of chassis, suspension, steering mechanisms, transmission, gear-box, differential systems, and electrical systems of various light and heavy automotive vehicles;



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Mechanical Engineering, VII-Semester

ME-75 Power Technology

Unit-I: Nuclear reactions and artificial radioactivity, introduction, energy and world, nuclear heat energy, nuclear fission and fusion and nuclear reactors, carbon dating, particle accelerators, International Thermonuclear Experimental Reactor, ITER, Large hadron collider, LHC, Radiation and materials, Biological effects of radiation, Nuclear propulsion, Radiation protection, Waste disposal

Unit-II: Solar energy, Introduction, an overview of thermal applications, Sun and geographic availability, Solar radiation, Thermal energy storage and utilization, Solar pond, Solar heaters, Solar collectors, Solar systems design, Passive heating systems, Economic analysis.

Unit-III: Wind technology- Introduction, Nature of wind, Place and direction, Energy in wind, wind data, Multi blade propeller type wind mill units and power calculations, Standardization, Conversion effectiveness to electrical and Mechanical Energy, Connectivity with grid, Wind machines classifications and applications, Turbines-Design and performance.

Unit-IV: Biomass technology, Introduction, Direct and indirect methods of bio mass resource utilization, Energy plantation, Biomass classification, Biomass general chemical thermodynamics, Combustion.

Unit-V: Hydrogen energy-Introduction, Collection and safety, Comparison with other power sources and effectiveness, Reforming and collection of hydrogen from water and hydrocarbons, Hydrogen fuel cells, Fuel cell efficiencies.

References:

1. Arthur Beiser, Concepts of modern Physics, TMH
2. R.L. Murray, Nuclear Energy, Pergamon Press.
3. S. P. Sukhatme, Solar Energy, TMH
4. J. F. Kreider- The solar heating design process, Mc Graw-Hill.
5. Philipp Kiamah-Power generation Handbook.
6. Wiley, Engineering Chemistry, Wiley



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Mechanical Engineering, VII-Semester

ME-76 CAD/CAM/CIM

The purpose of this laboratory is to provide the complete practical exposure of Computer aided design software tools such as Solid works, CATIA, Unigraphics etc., Computer assisted manufacturing processes such as CNC Turning, CNC Milling, CNC Drilling etc as well Computer integrated manufacturing (i.e. demonstrating remotely over the internet the operations of actual CIM cell established in the dept of Institute /industry) to the students so that they will become industry ready just after completing their graduation.

Suggested list Experiments;

1. 2D and 3D modeling on CAD software
2. Use of CAM software for writing CNC programs
3. Study of automatic and semi automatic control system and writing the electrical analogy.
4. Production & layout for GT for group of jobs to be manufactured
5. A case study / tutorial using CAPP Software
6. Writing M & G codes for given operations.
7. Robot and AGV programming
8. Modelling and simulation of computer integrated manufacturing system'
9. Modelling,offline manual part programming and simulation of the operation of 3 axis CNC milling machine
10. Programming and operation of a 5 axis robot Manipulator
11. Remote monitoring and operation of Computer integrated manufacturing system
12. To write the part program for any component (stepped cylindrical rod) . Assuming the work piece is Aluminum and the speed is 1200 rpm, feed 20 mm/min and maximum depth of cut is 1 mm.



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Mechanical Engineering, VII-Semester

ME-76 CAD/CAM/CIM

- a. With Canned cycle
- b. Without Canned cycle.

Evaluation:

Evaluation will be continuous and integral part of the class as well as through external assessment (Viva/voce)



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Mechanical Engineering, VII-Semester

ME-77 Industrial Training

Objective of Industrial Training

The objective of undertaking industrial training is to provide work experience so that student's engineering knowledge is enhanced and employment prospects are improved. The student should take this course as a window to the real World and should try to learn as much as possible from real life experiences by involving and interacting with industry staff. Industrial training also provides an opportunity to students to select an engineering problem and possibly an industry guide for their Major Project in final semester.

Scheme of Studies:

Duration: Minimum 2 weeks in summer break after VI semester, assessment to be done in VII Semester. Marks of various components in industry should be awarded by the I/c of training in Industry but in special circumstances if not awarded by the industry then faculty in charge /T.P.O. will give the marks. During training students will prepare a first draft of training report in consultation with section in charge. After training they will prepare final draft with the help of T.P.O. /Faculty of the Institute. Then they will present a seminar on their training and they will face viva-voce on training in the Institute.

Learning through Industrial Training

During industrial training students must observe following to enrich their learning:

- ☐ Industrial environment and work culture.
- ☐ Organizational structure and inter personal communication.
- ☐ Machines/equipment/instrument-their working and specifications.
- ☐ Product development procedure and phases.
- ☐ Project Planning, monitoring and control.
- ☐ Quality control and assurance.
- ☐ Maintenance system
- ☐ Costing system
- ☐ Stores and purchase systems.
- ☐ Layout of Computer/EDP/MIS centers.



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ME-77 Industrial Training

• **Roles and responsibilities of different categories of personnel.**

- ☐ Customer services.
 - ☐ Problems related to various areas of work etc.
- Students are supposed to acquire the knowledge on above by-
- ☐ Direct Observations without disturbing personnel at work.
 - ☐ Interaction with officials at the workplace in free/ tea time
 - ☐ Study of Literature at the workplace (e.g. User Manual, standards, processes, schedules, etc.)
 - ☐ “Hand’s on” experience
 - ☐ Undertaking/assisting project work.
 - ☐ Solving problems at the work place.
 - ☐ Presenting a seminar
 - ☐ Participating in group meeting/discussion.
 - ☐ Gathering primary and secondary data/information through various sources, storage, retrieval and analysis of the gathered data.
 - ☐ Assisting official and managers in their working
 - ☐ Undertaking a short action research work.
 - ☐ Consulting current technical journals and periodicals in the library.
 - ☐ Discussion with peers.

Daily Diary- Industrial Training

Name of the Trainee----- College -----
Industry / work place ----- Week No-----
Department /Section ----- Date -----

Dates Brief of observations made, work done, problem/project undertaken, discussion held, literature consulted etc



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ME-77 Industrial Training

Signature of Supervisor Signature of Trainee Signature of Official in (TPO/Faculty) charge for
Trg. In Industry.

Supervision of Industrial Training

Faculty and TPO are supposed to plan industrial training in such a manner that students get exposure on most of the above area in the field. One faculty member or TPO will plan industrial training of students in consultation with training manager of the industry (work place) as per the predefined objectives of training. Monitoring visits will be made by training and placement officer/faculty in-charge for the group of students, of the college during training.

Guidance to the faculty / TPO for Planning and implementing the Industrial Training

Keeping in view the need of the contents, the industrial training program, which is spread to minimum 2 weeks duration, has to be designed in consultation with the authorities of the work place; Following are some of the salient points:

- ☐ Spelling out the objectives of the industrial training in behavioral terms and same is informed in advance to the 1) students, 2) authorities of the work place and 3) supervising faculty members.
- ☐ Discussing and preparing students for the training for which meetings with the students has to be planned.
- ☐ Meeting with industrial personnel and orienting them regarding the objective of the training and the expectations of the program.
- ☐ Correspondence with the authorities of the work place.
- ☐ Orientation classes for students on how to make the training most beneficial- monitoring daily diary, writing weekly reports, how to interact with various categories of industrial personnel, how to behave and undertake responsibilities, how to gather information from the workplace, ethics etc.
- ☐ Guiding students to make individual plans (week wise/ day wise) to undertake industrial training.,
- ☐ Developing a system of maintaining training records, by teachers for every batch of students for convenient retrieval



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Mechanical Engineering, VII-Semester

ME-77 Industrial Training

Action plan for planning stages at the Institutional Level

S.No. Activity Commencing Week Finishing week Remark

1. Meeting with Principal
2. Meeting with colleagues
3. Correspondence with work place(Industry concerned)
4. Meeting with authorities of work place
5. Orientation of students for industry training
6. Scrutinizing individual training plan of students.
7. Commencement of individual training
8. First monitoring of industrial training
9. Second monitoring of industrial training
10. Finalization of Training report
11. Evaluation of performance at industry level
12. Evaluation of Industry Program in the Institutions.